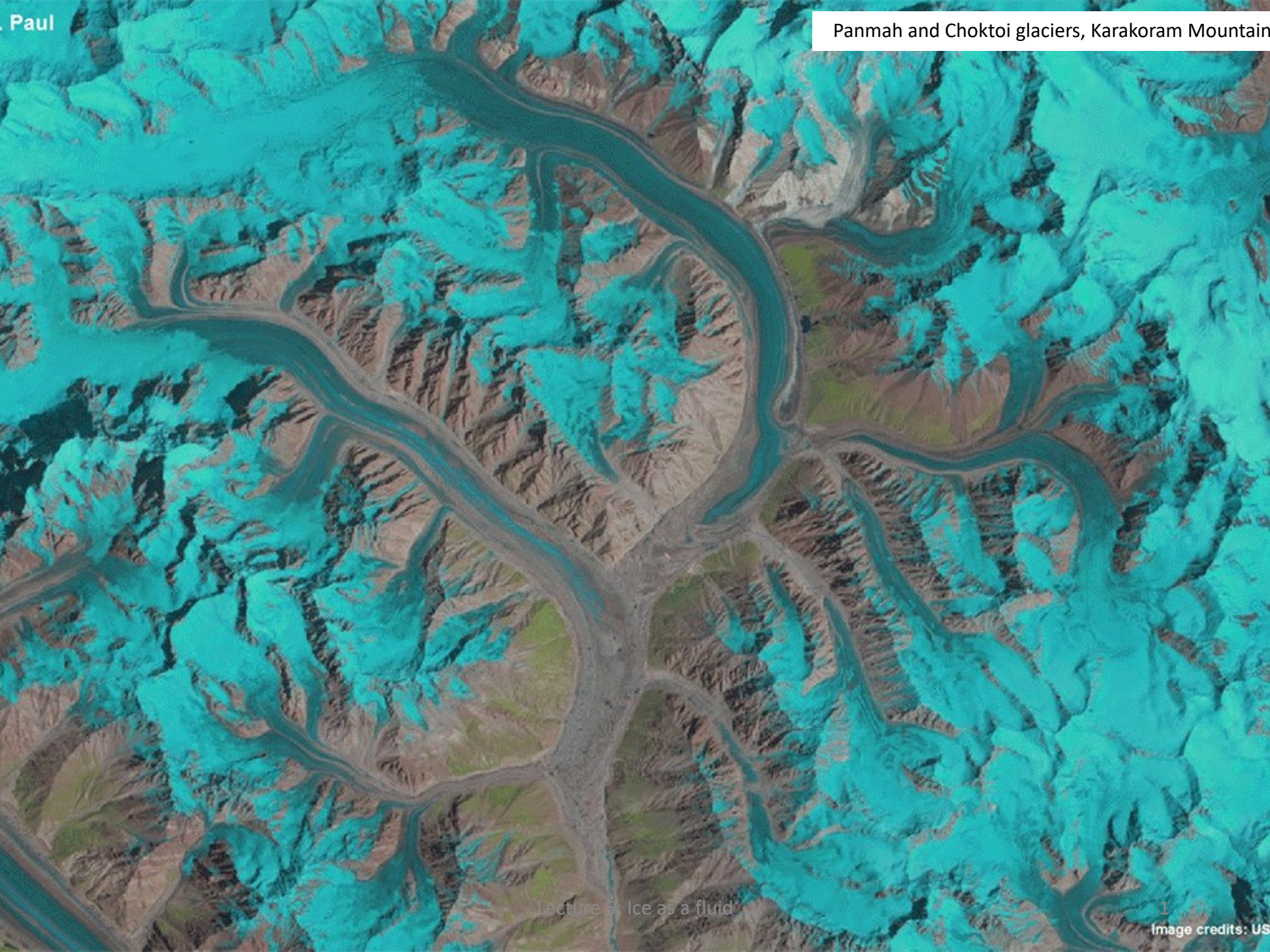




Lecture 5:

Ice as a nonlinear fluid

Deformation

Deformation mechanisms

A basic flow law for ice

The stress *tensor*

Stress balance equations

Strain rates and velocity gradients.

Effective stress, deviatoric stress and Glen's flow law

Deformation

Elastic deformation

$$\epsilon = \frac{\tau}{E}$$

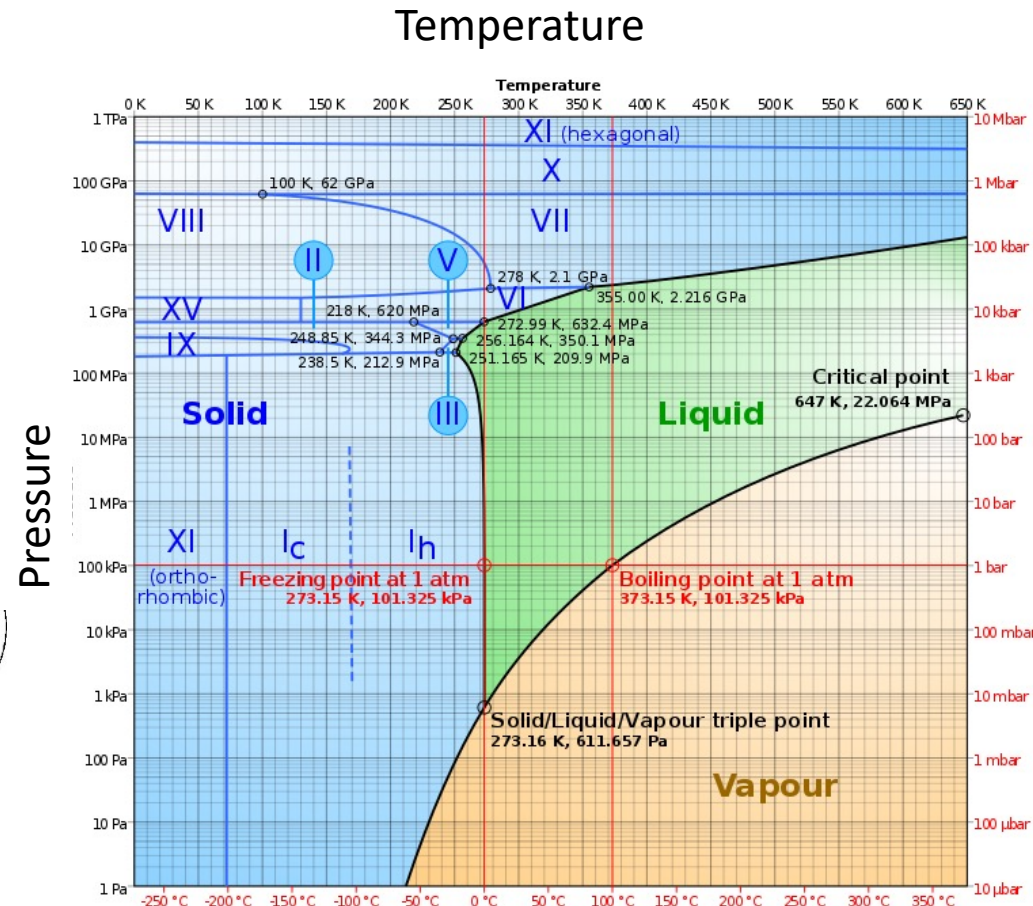
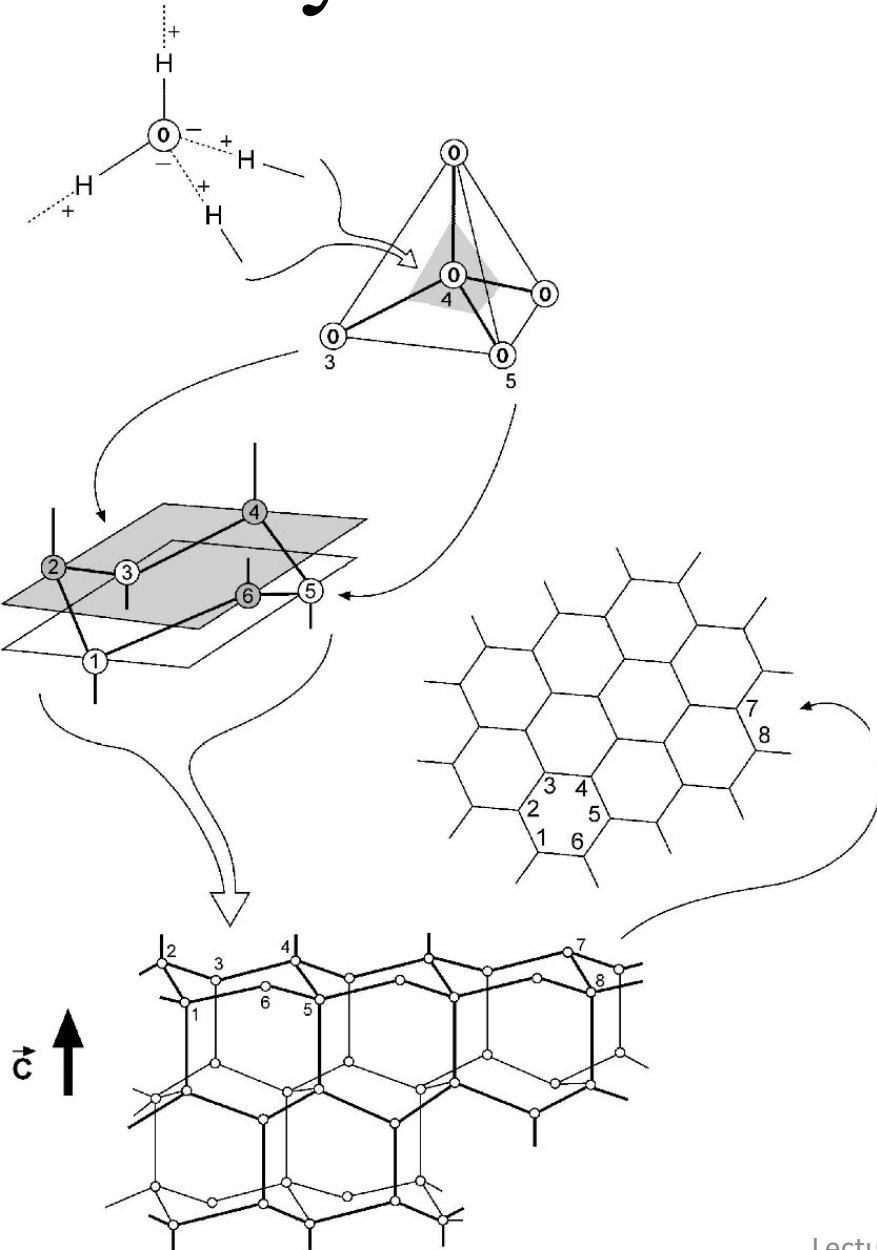
ϵ = strain [-]
 τ = stress [Pa]
 E = elastic modulus [Pa]

Viscous deformation

$$\dot{\epsilon} = \frac{\tau}{\mu}$$

$\dot{\epsilon}$ = strain rate [1/s]
 τ = stress [Pa]
 μ = viscosity [Pa s]

Ice crystal structure



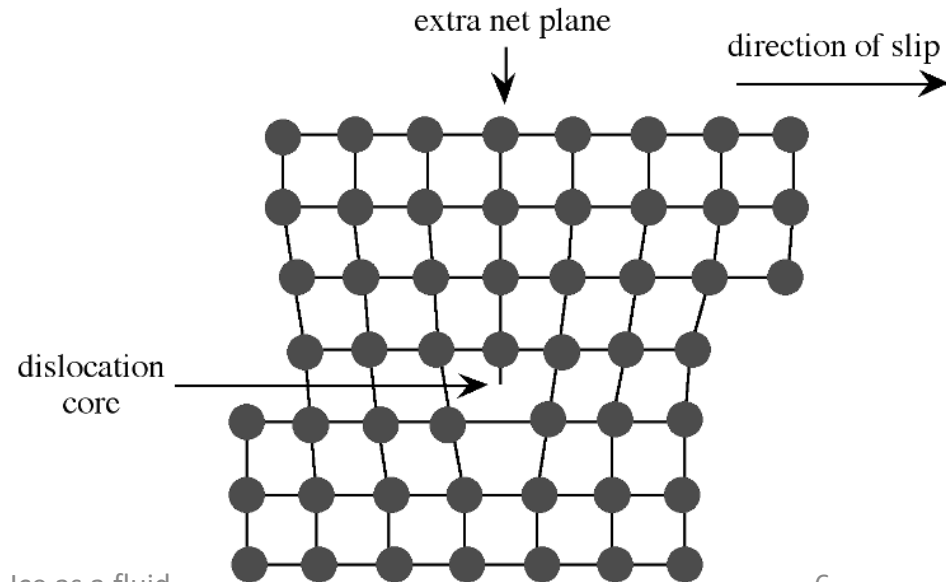
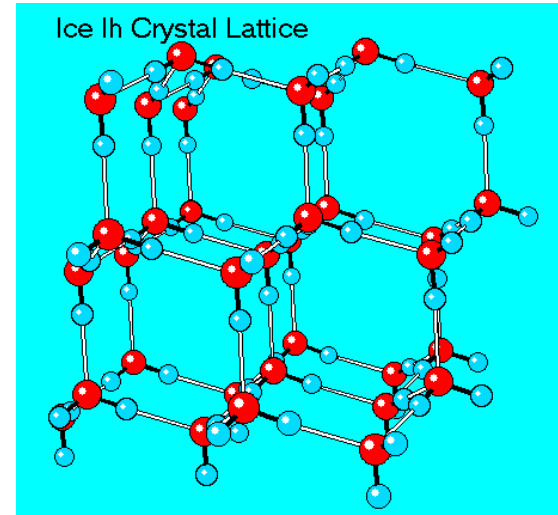
Crystals and Grains



6 cm

Theory: Deformation of single grains

- Dislocation creep
 - Dislocations in lattice move through the crystal.
 - Highly anisotropic
- Depends sensitively on temperature.



Theory: deformation of polycrystalline ice

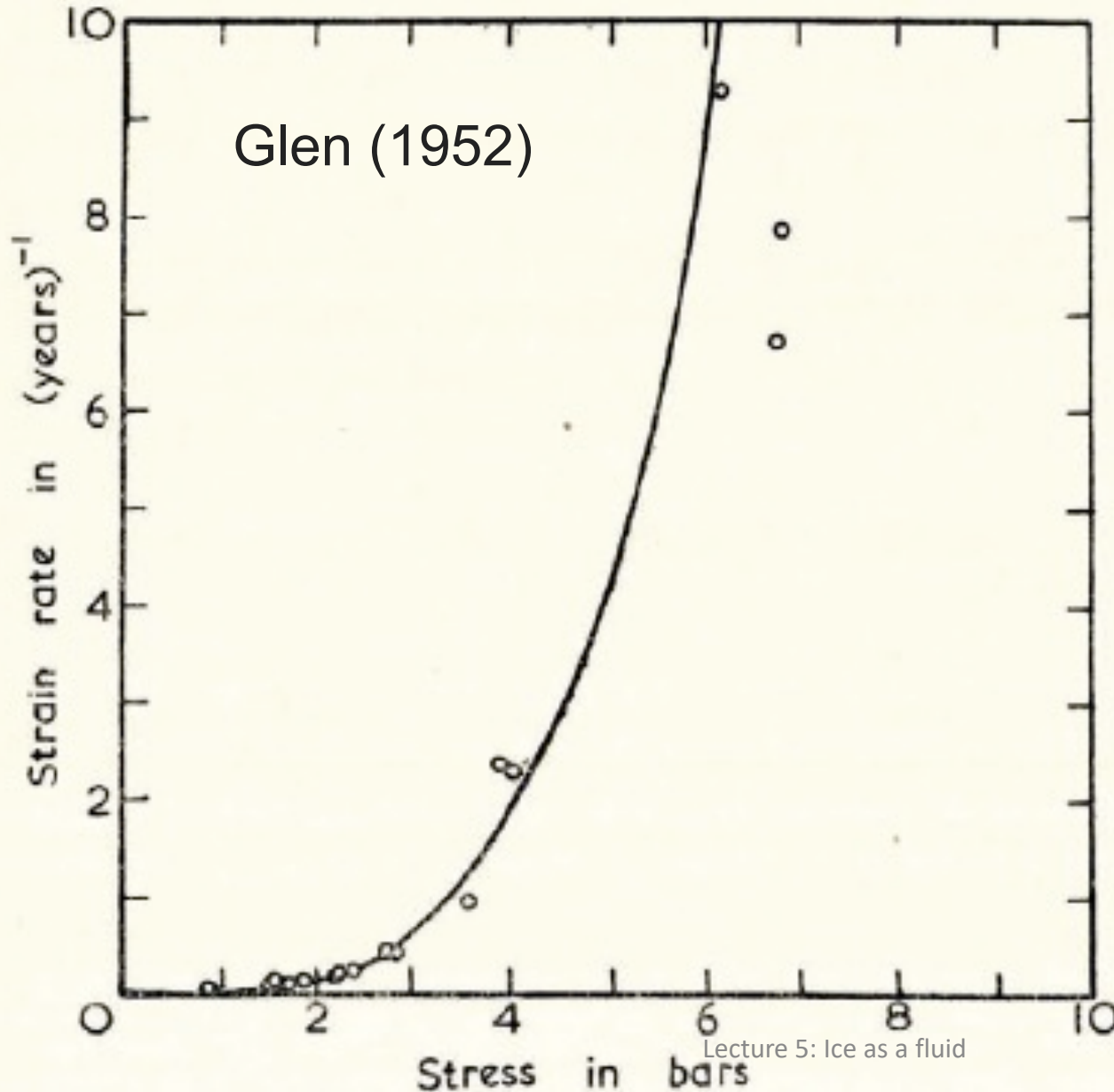
- Grain orientation is non-uniform.
- Individual grains deform via dislocation creep ($n \approx 3-4$).
- Neighboring grains deform at very different rates.
- “Accommodated” by grain boundary mechanisms, such as grain boundary sliding ($n \approx 1.8$).

$$\dot{\epsilon} = A\tau^n$$

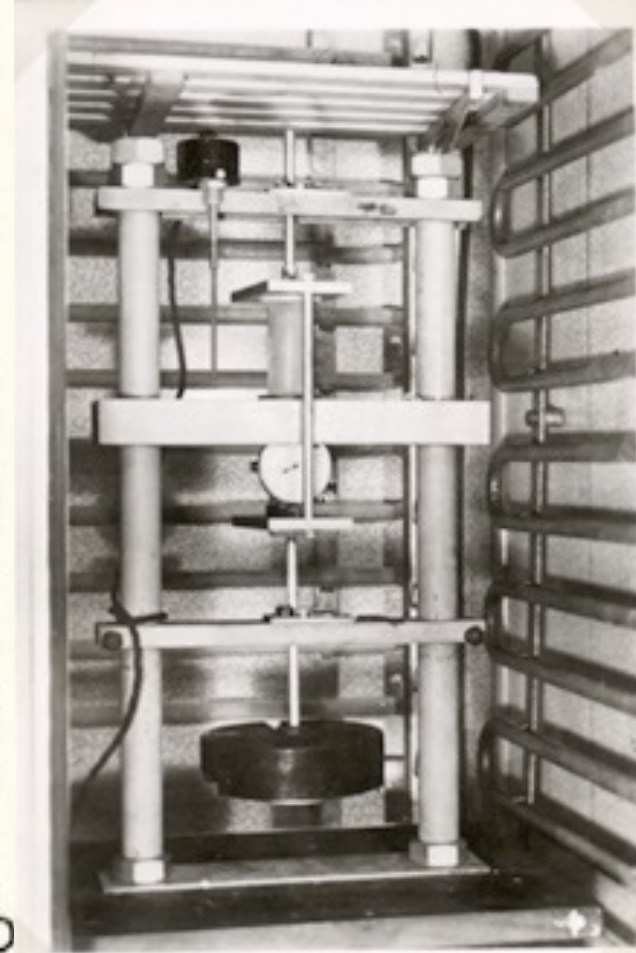


Observational evidence for $n > 1$

Glen (1952)



$$\dot{\epsilon} = A\tau^n$$



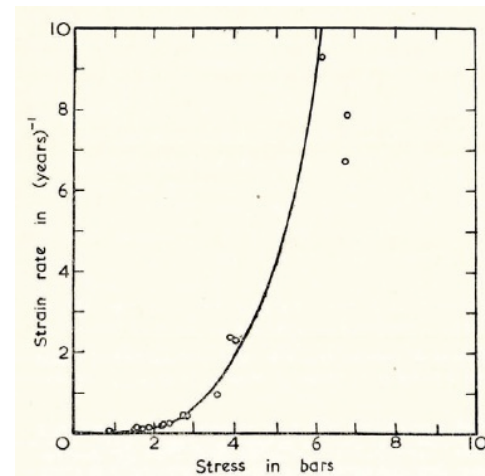
Observational evidence for $n > 1$

- Lab experiments explore the different stress and strain regimes with varying success.
- Glen (1952) is the canonical experiment.

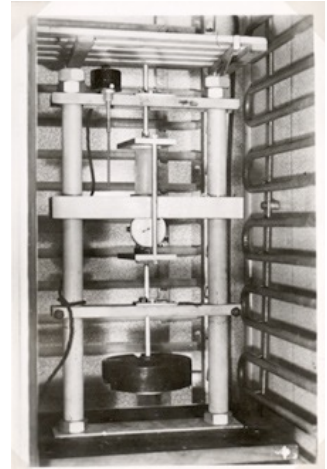
--> Glen's flow law

- Field observations:
 - Ice shelf spreading rates
 - Borehole closure rates
 - Vertical velocity profiles

The consensus has been that $n = 3$ is a good approximation.



Glen (1952)

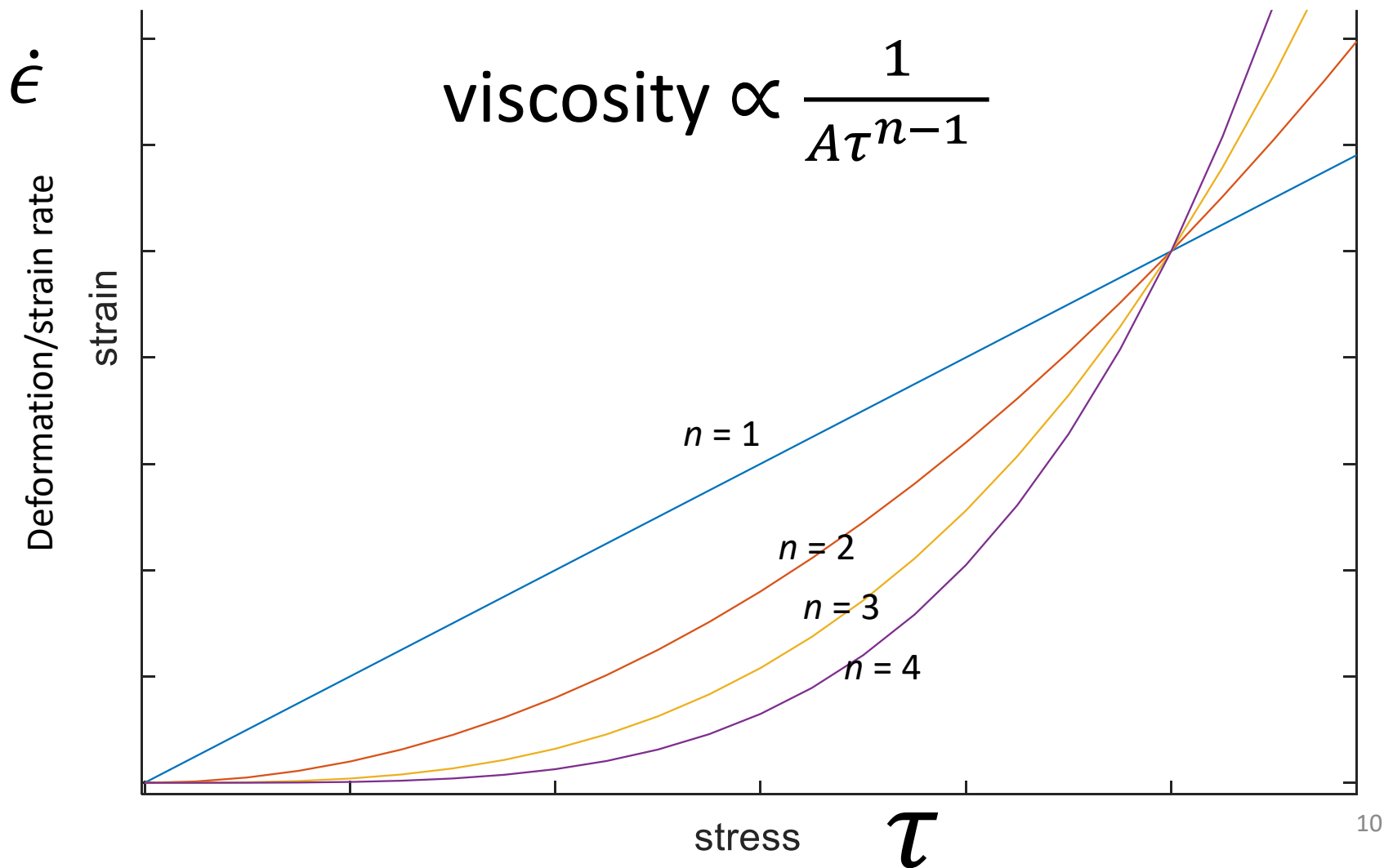


$$\dot{\epsilon} = A\tau^n$$

$$\dot{\epsilon} = \frac{\tau}{\mu}$$

$$\dot{\epsilon} = A\tau^n$$

$$\text{viscosity} \propto \frac{1}{A\tau^{n-1}}$$



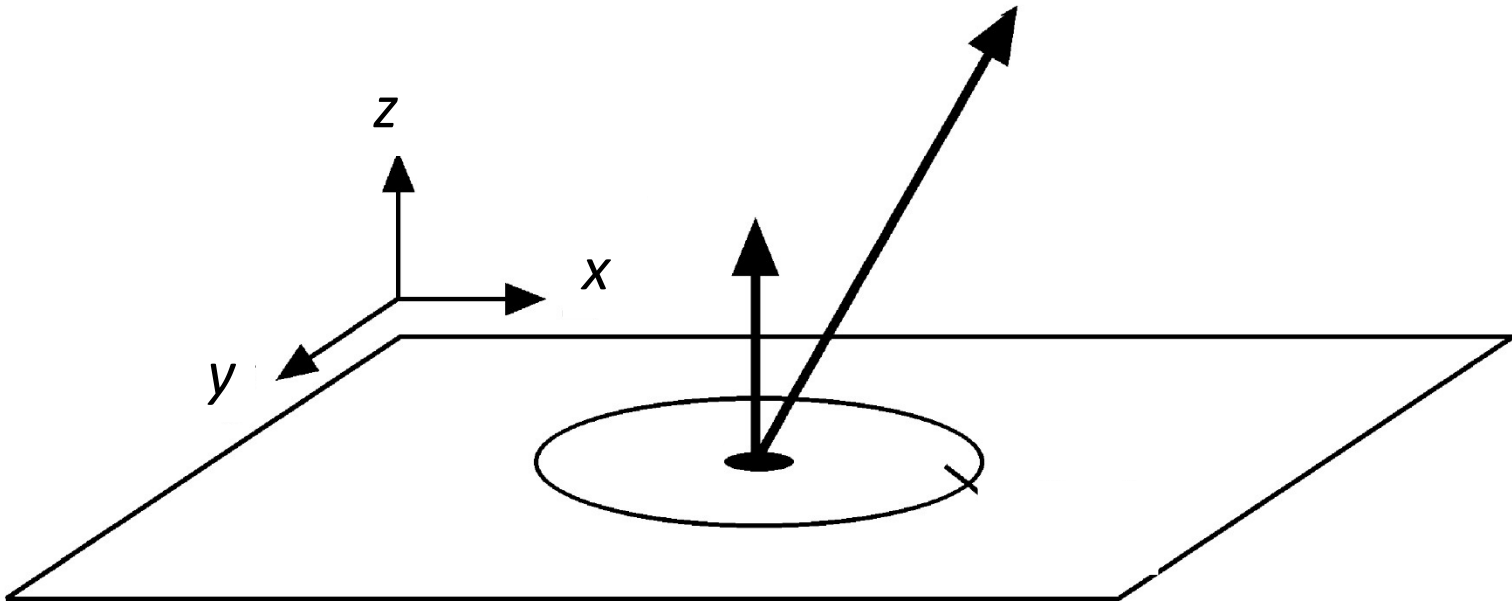
Glen's flow law

$$\dot{\epsilon} = A\tau^n$$

$$\text{viscosity} \propto \frac{1}{A\tau^{n-1}}$$

- A varies over several orders of magnitude in glaciers and ice sheets.
 - A increases with:
 - Temperature
 - Pressure
 - Water content
 - Impurities
 - Grain size
 - Preferred crystal orientation fabric.
- Tightly coupled properties
--> difficult to untangle the effects of each.

Stress



Stress: force per unit area [units: Pa, i.e. N m^{-2}]

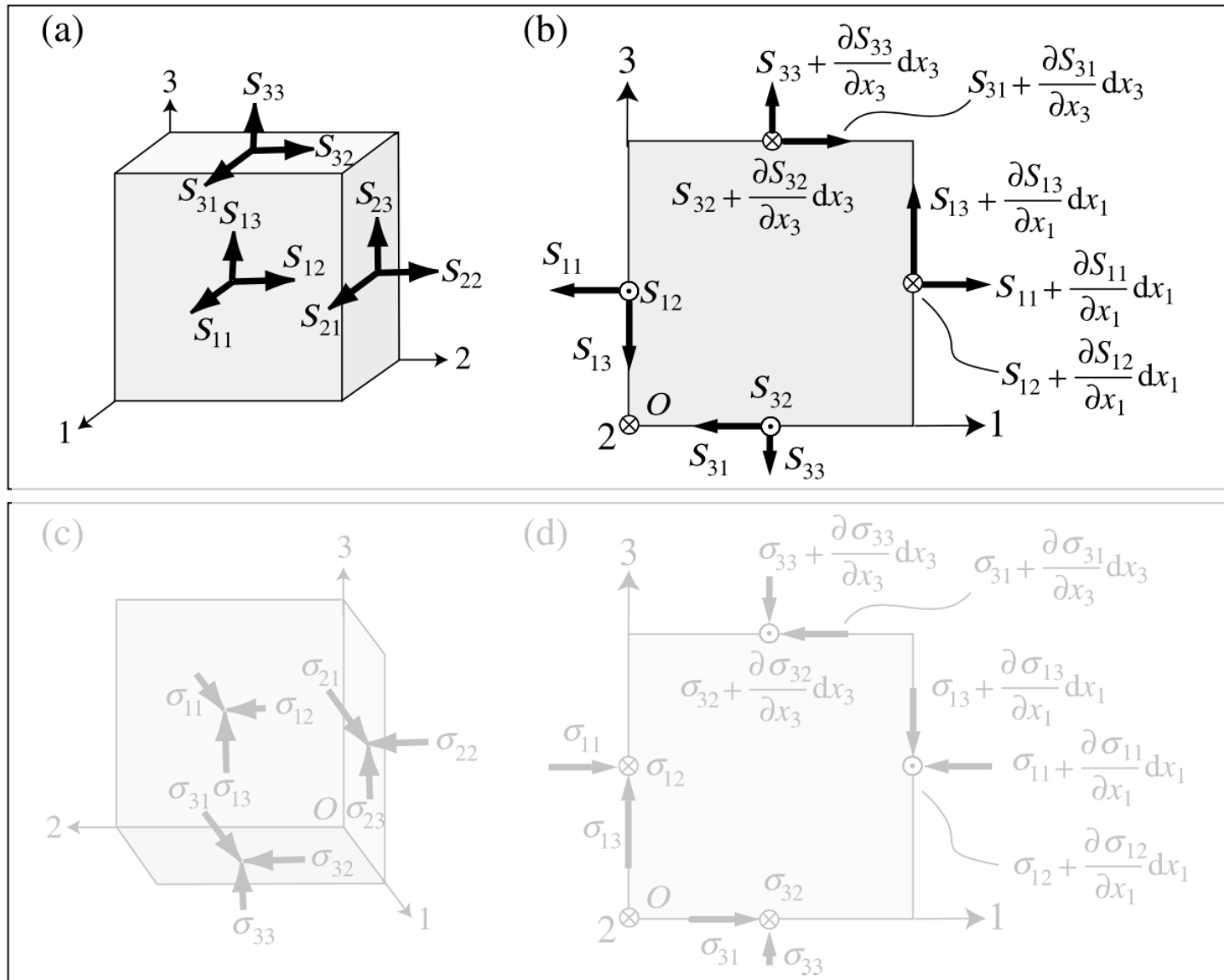
Has components perpendicular to this plane

$$\sigma_{zz}$$

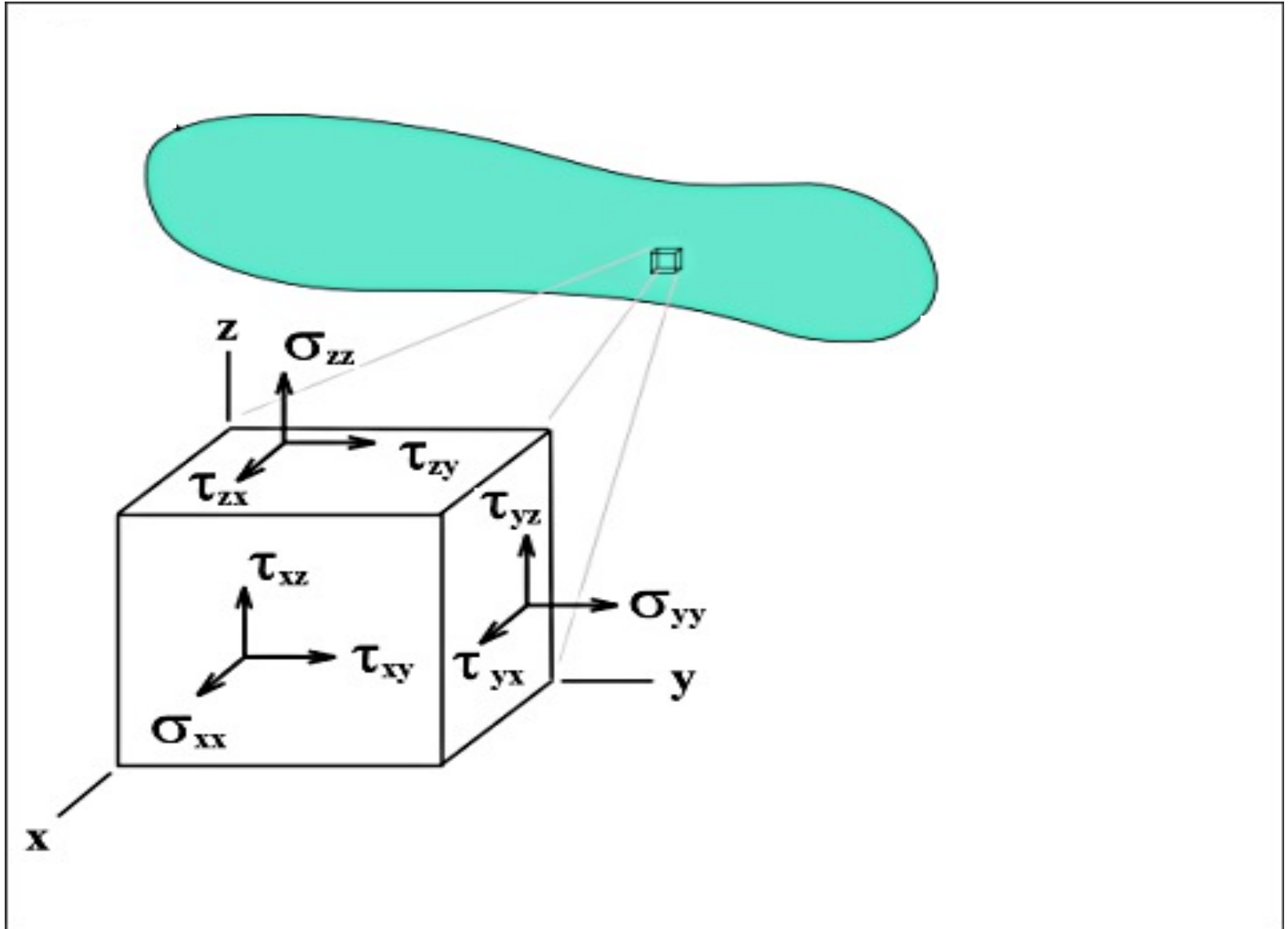
and in the plane

$$\tau_{zy} \quad \& \quad \tau_{zx}$$

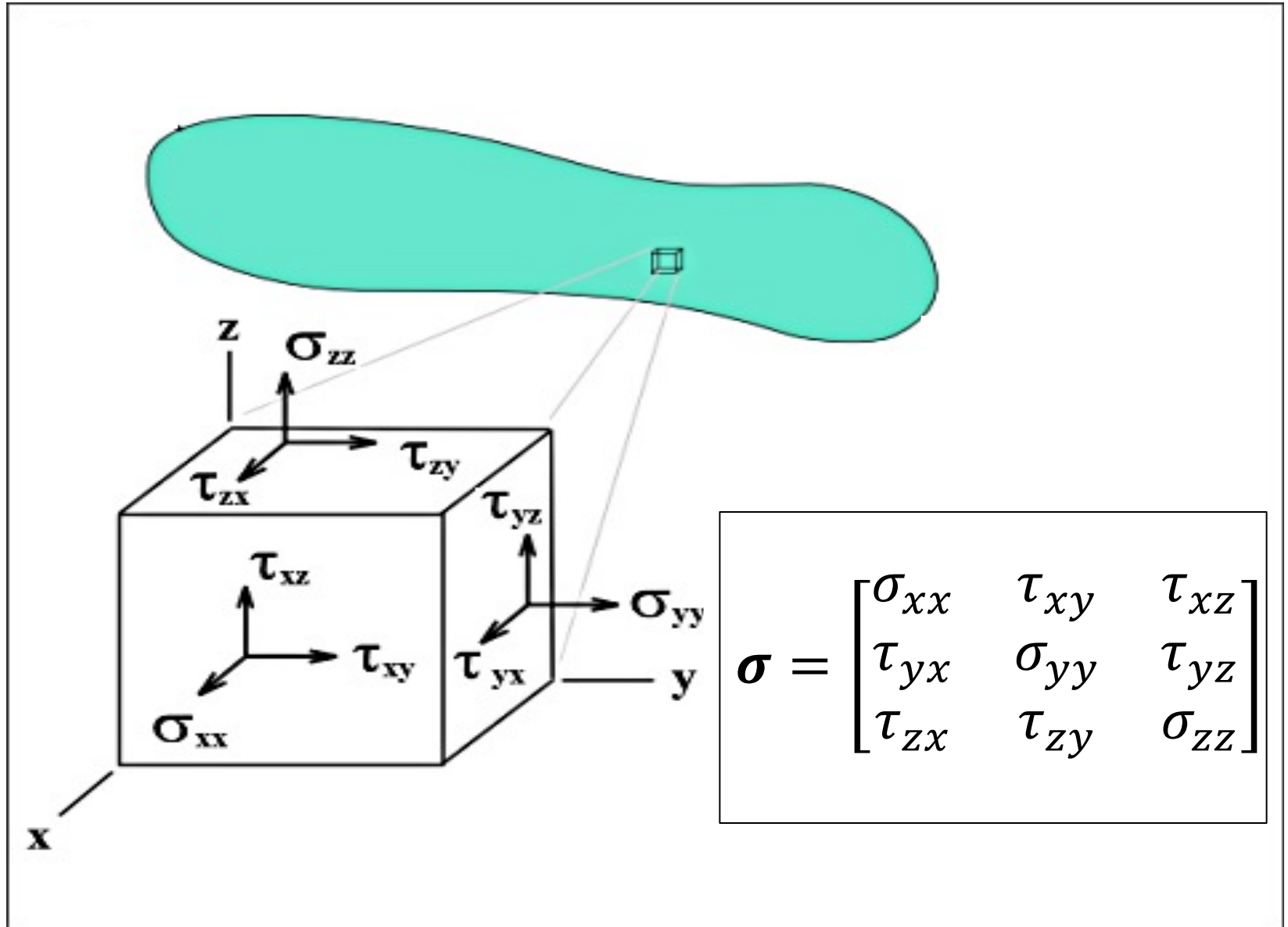
Sign convention in glaciology:



Stress tensor



Stress tensor



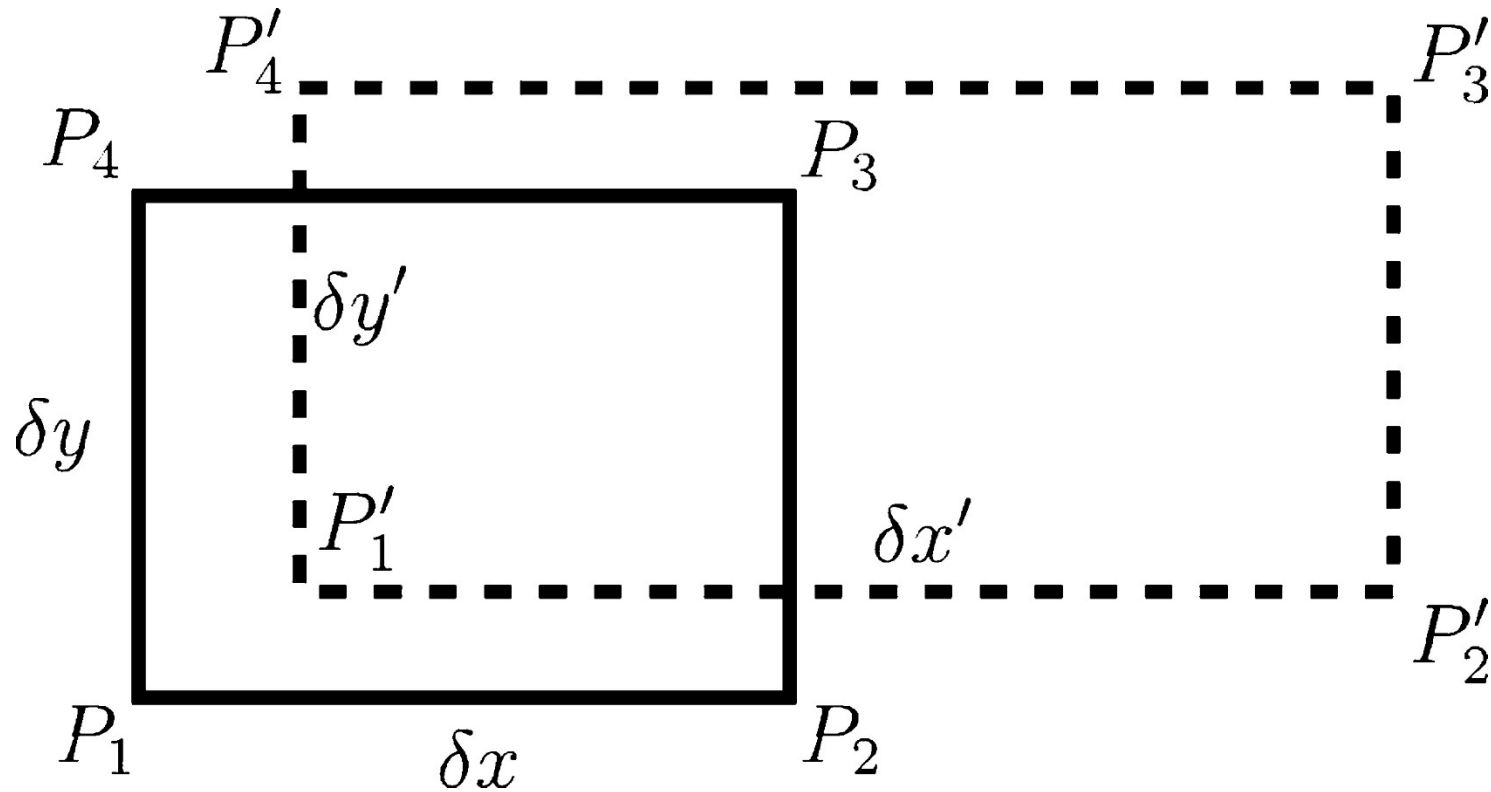
Stress-balance equations

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} = 0 \quad \text{x-direction}$$

$$\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} = 0 \quad \text{y-direction}$$

$$\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} = \rho g \quad \text{z-direction}$$

Normal strain and velocity gradients



Strain rate

$$\dot{\epsilon}_{ij} = \frac{1}{2} \left[\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j} \right]$$

$$\dot{\epsilon} = \begin{bmatrix} \dot{\epsilon}_{xx} & \dot{\epsilon}_{xy} & \dot{\epsilon}_{xz} \\ \dot{\epsilon}_{yx} & \dot{\epsilon}_{yy} & \dot{\epsilon}_{yz} \\ \dot{\epsilon}_{zx} & \dot{\epsilon}_{zy} & \dot{\epsilon}_{zz} \end{bmatrix}$$

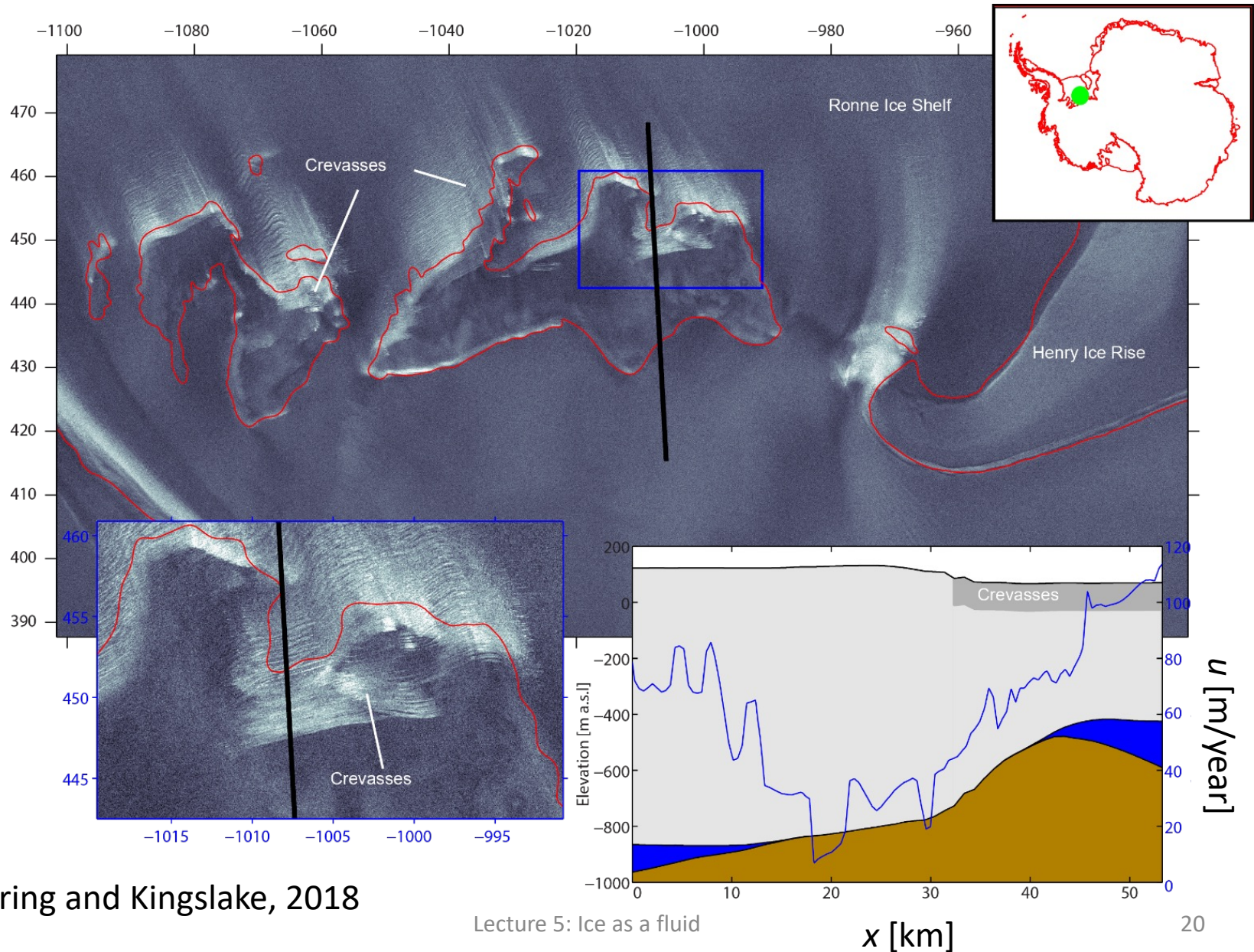
Strain rate

$$\epsilon_{ij}^{\dot{}} = \frac{1}{2} \left[\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j} \right]$$

For example:

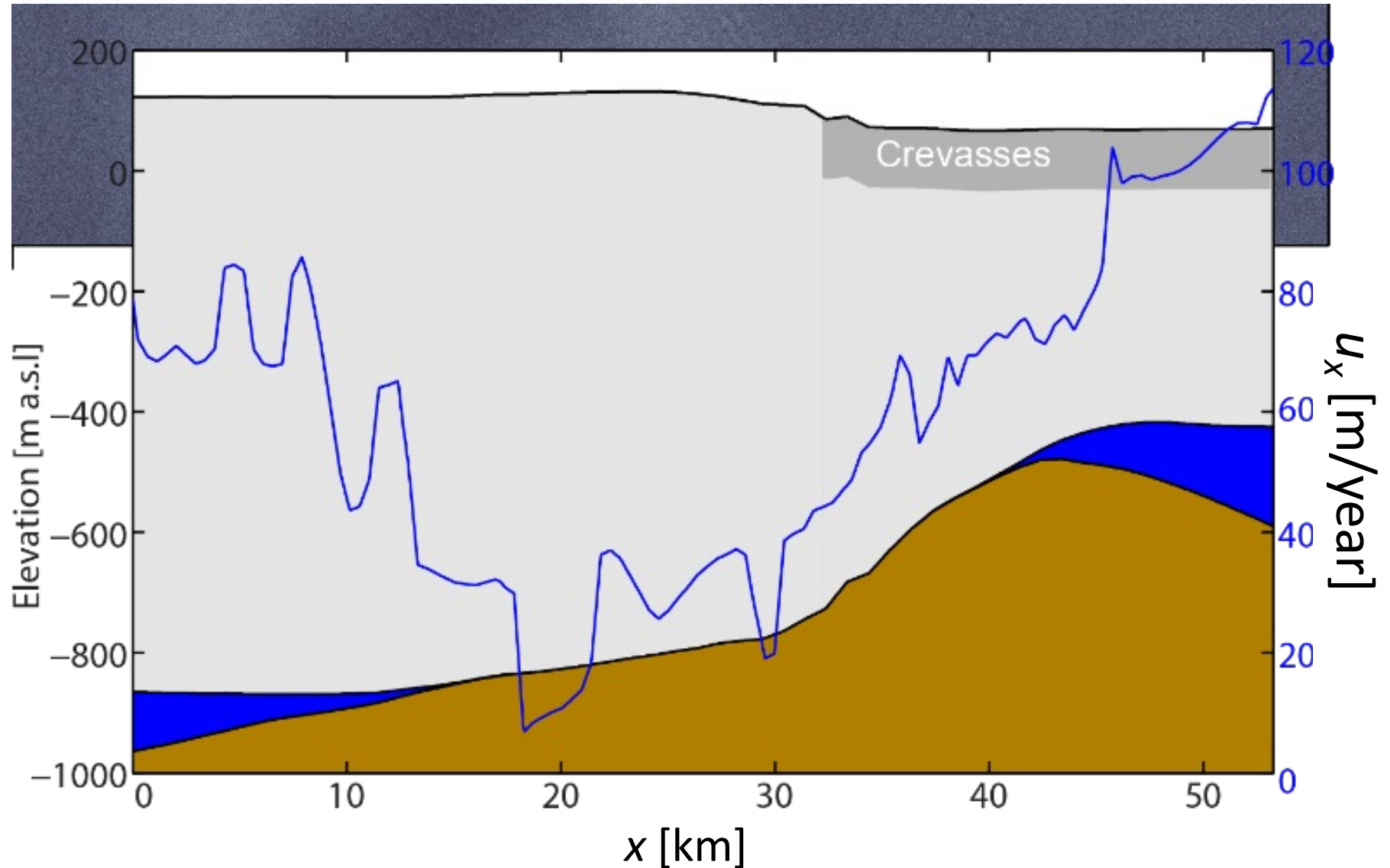
$$\epsilon_{xx}^{\dot{}} = \frac{\partial u_x}{\partial x} \quad \text{or} \quad \epsilon_{xz}^{\dot{}} = \frac{1}{2} \left[\frac{\partial u_x}{\partial z} + \frac{\partial u_z}{\partial x} \right]$$

A velocity profile from WAIS



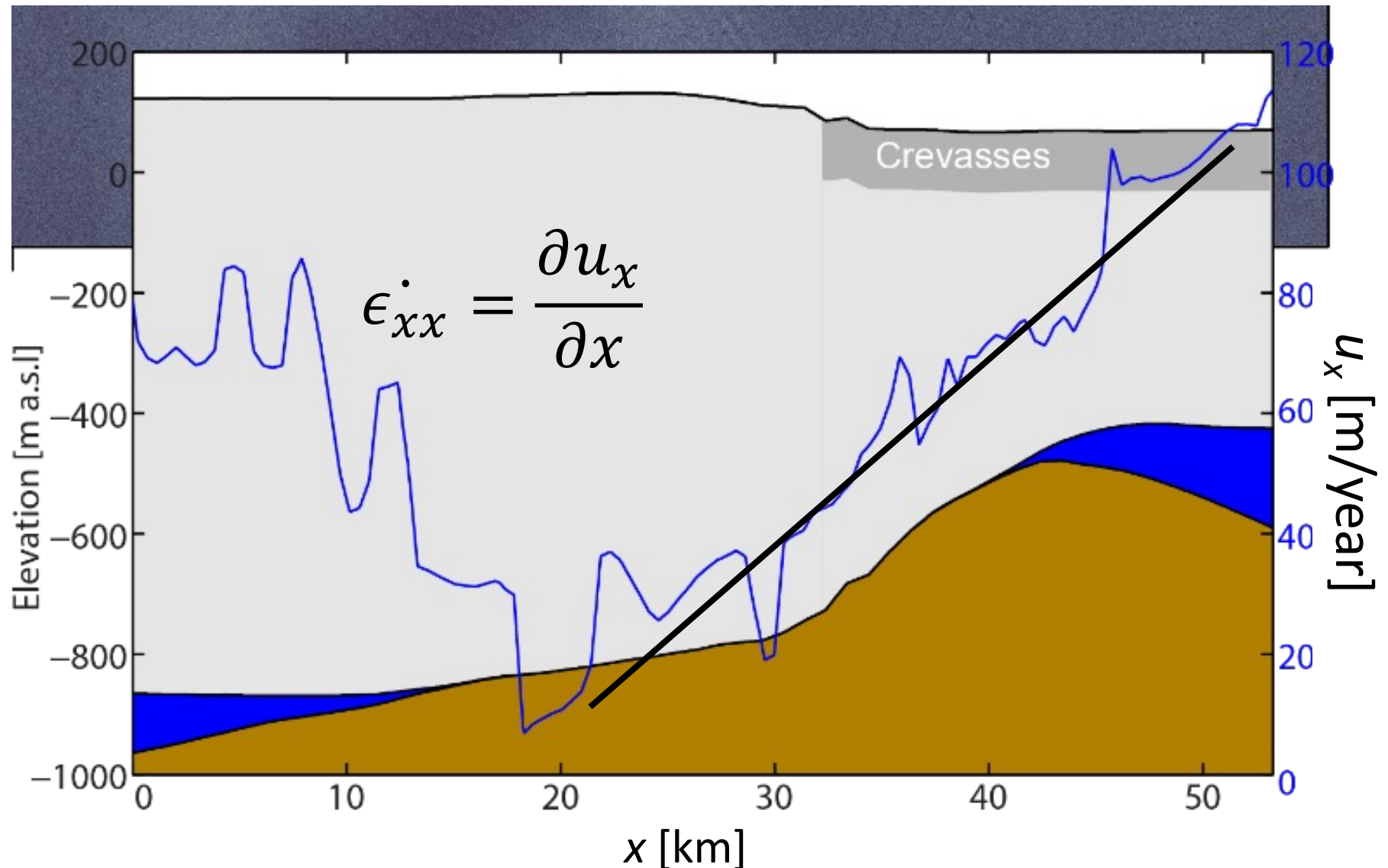
Wearing and Kingslake, 2018

A velocity profile from WAIS



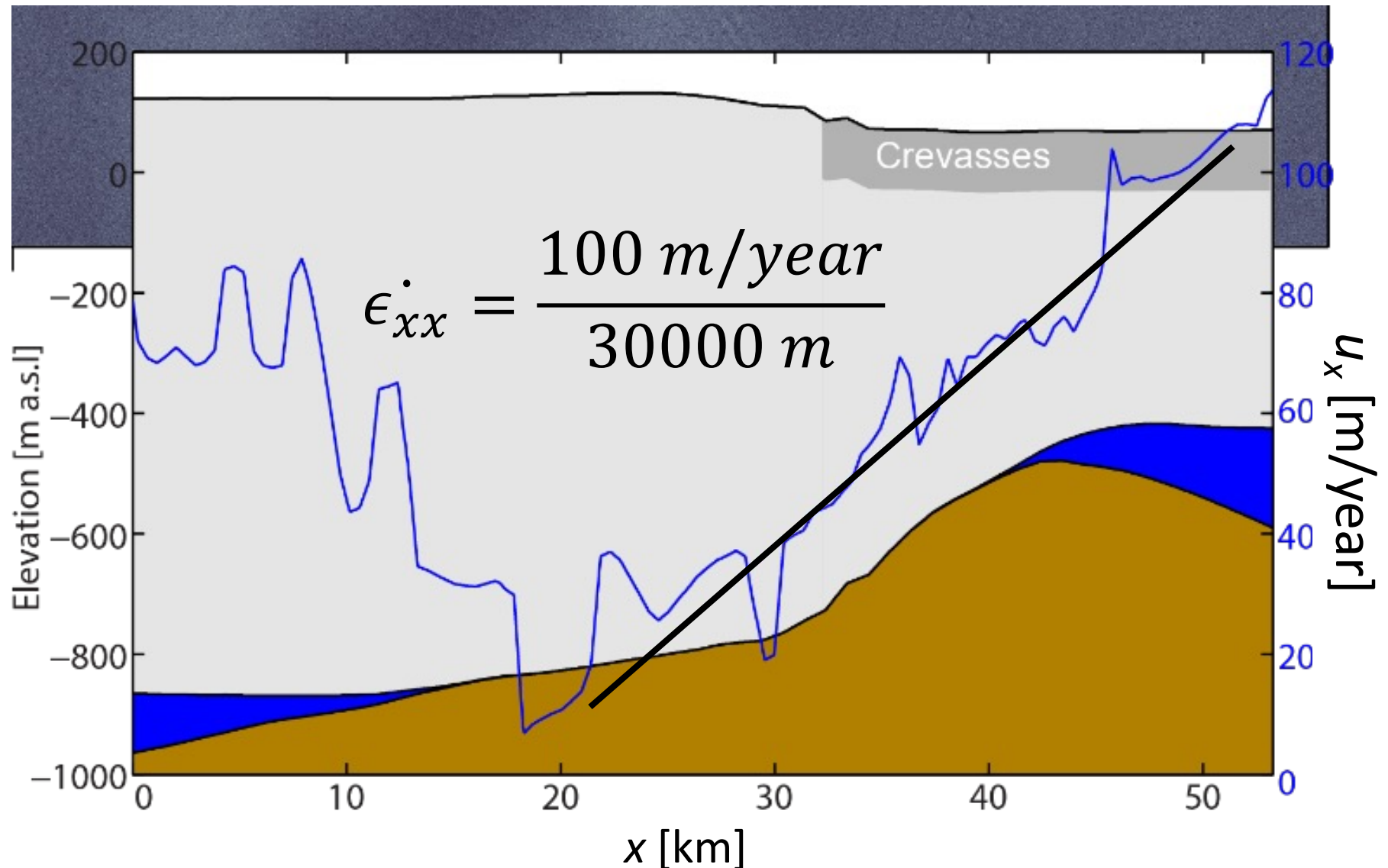
Wearing and Kingslake, 2018

A velocity profile from WAIS



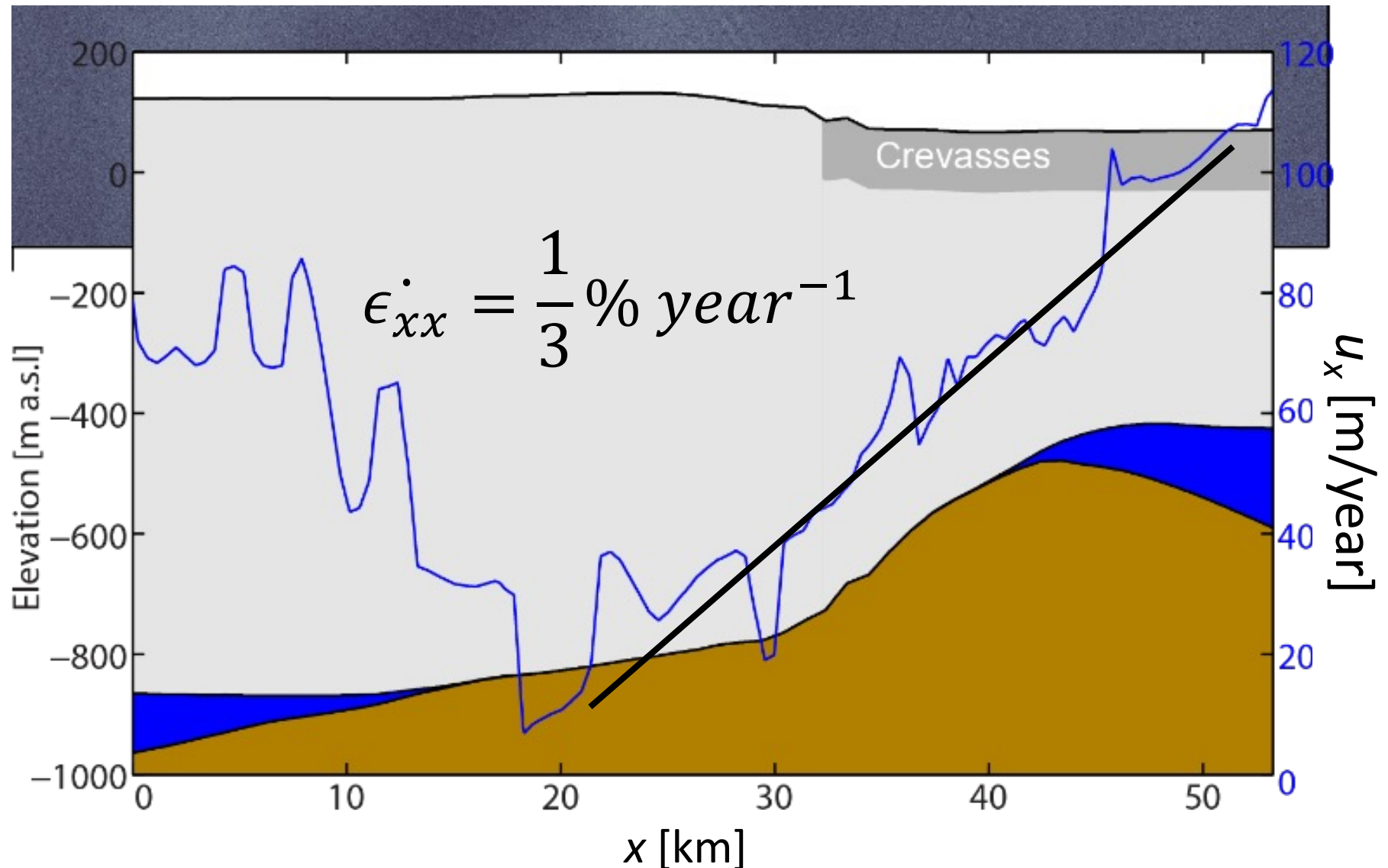
Wearing and Kingslake, 2018

A velocity profile from WAIS



Wearing and Kingslake, 2018

A velocity profile from WAIS



Wearing and Kingslake, 2018

How to relate these more rigorous definitions of stress and strain?

Deviatoric stress

Because ice is incompressible, it is the 'deviatoric stress' that we care about.

$$\sigma = \tau + \sigma_m$$

Total stress

Deviatoric stress

- Mean normal stress
- Hydrostatic stress
- Isotropic stress
- Volume stress

$$\sigma_m = \frac{1}{3} [\sigma_{xx} + \sigma_{yy} + \sigma_{zz}] I$$

$$\tau = \begin{bmatrix} \sigma_{xx} - \sigma_m & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_{yy} - \sigma_m & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_{zz} - \sigma_m \end{bmatrix}$$

Deviatoric stress is the total stress minus the mean normal stress.

Deviatoric stress

Because ice is incompressible, it is the 'deviatoric stress' that we care about.

$$\sigma = \tau + \sigma_m$$

Total stress

Deviatoric stress

- Mean normal stress
- Hydrostatic stress
- Isotropic stress
- Volume stress

$$\sigma_m = \frac{1}{3} [\sigma_{xx} + \sigma_{yy} + \sigma_{zz}] I$$

$$\tau = \begin{bmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{bmatrix}$$

Effective stress

$$\boldsymbol{\tau} = \begin{bmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{bmatrix}$$

- The “*second invariant of the stress tensor*” is called the effective stress in glaciology.
(we wont derive it here)

$$\tau_E^2 = \frac{1}{2} [\tau_{xx}^2 + \tau_{yy}^2 + \tau_{zz}^2] + \tau_{xz}^2 + \tau_{xy}^2 + \tau_{yz}^2$$

It is a scalar and is independent of the axes you use.

Back to Glen's flow law....

$$\epsilon_{ij}^{\dot{}} = A \tau_E^{n-1} \tau_{ij}$$

$$\tau_E^2 = \frac{1}{2} [\tau_{xx}^2 + \tau_{yy}^2 + \tau_{zz}^2] + \tau_{xz}^2 + \tau_{xy}^2 + \tau_{yz}^2$$

We will be using this relation, the stress balance equations and the idea of deviatoric stress next week to derive a simple ice sheet model.

Summary

- Ice deforms through various mechanisms, dislocation creep in the basal plane dominates.
- Rheology is the relationship between stress and strain.
- Nonlinear rheology of ice is usually approximated by Glen's flow law with $n = 3$.
- Stress state is described by the “stress tensor”.
- Effective stress is a measure of the amount of stress at a particular location.
- Strain is also a tensor. Its elements are spatial derivatives of the velocity field.

References

Gillet-Chaulet, F., Hindmarsh, R.C., Corr, H.F., King, E.C. and Jenkins, A., 2011. In-situ quantification of ice rheology and direct measurement of the Raymond Effect at Summit, Greenland using a phase-sensitive radar. *Geophysical Research Letters*, 38(24).

Glen, J.W., 1952. Experiments on the deformation of ice. *Journal of Glaciology*, 2(12), pp.111-114.

Wearing, M.G. and Kingslake, J., 2019. Holocene Formation of Henry Ice Rise, West Antarctica, Inferred from Ice-Penetrating Radar. *Journal of Geophysical Research: Earth Surface*, 124(8), pp.2224-2240.